Event-Aligned SVD for Data-Driven Pattern Kernel Discovery

Mathematical Specification for the NAT Convolver Pipeline

NAT Quantitative Research Platform

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Abstract

We present a complete mathematical specification for a data-driven pattern detection system applied to OHLCV candle data from Hyperliquid perpetual futures. The core pipeline defines market events analytically (breakouts, turtle-soup false breakouts, bull/bear traps), extracts aligned windows of candle data around those events, applies Singular Value Decomposition to discover characteristic multi-candle shapes, filters the discovered components by forward-return Information Coefficient, and deploys the surviving kernels as a real-time convolver. The approach combines the interpretability of analytical event definitions with the statistical efficiency of data-driven basis discovery, with three explicit safeguards against overfitting. The document is self-contained: a reader familiar with linear algebra and basic time-series statistics should be able to implement the full pipeline from this specification.

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1 Notation Table

Symbol	Type / Range	Meaning
t	$t \in \mathbb{N}$	Discrete candle time index
O_t, H_t, L_t, C_t	$\mathbb{R}_{>0}$	Open, High, Low, Close prices at bar t
V_t	$\mathbb{R}_{\geq 0}$	Traded volume at bar t
b_t	$\mathbb{R}$	Candle body: $C_t - O_t$
u_t	$\mathbb{R}_{\geq 0}$	Upper wick
l_t	$\mathbb{R}_{\geq 0}$	Lower wick
v_t	$\mathbb{R}_{\geq 0}$	Volume channel (to be normalised)
W	$W \in \mathbb{N}$	Window width in candles
N	$N \in \mathbb{N}$	Lookback for event detection
K	$K \in \mathbb{N}$	Confirmation lag (trap events)
α	$\alpha > 0$	ATR multiplier for trap threshold
t_e	$t_e \in \mathbb{N}$	Detection time of event e
n_e	$n_e \in \mathbb{N}$	Number of events of type e
$\mathcal{E}$	set	Set of event types (6 elements)
$\mathcal{C}$	$\{b, u, l, v\}$	Set of signal channels
τ	$\tau \in \{0, \ldots, W-1\}$	Within-window time index
$\bar{\tau}$	$\bar{\tau} \in [0, 1]$	Normalised window coordinate $\tau/(W-1)$
$\tilde{x}_i^{(c)}(\tau)$	$\mathbb{R}$	Normalised channel c, event i, lag τ
$\mathbf{X}^{(e,c)}$	$\mathbb{R}^{n_e \times W}$	Event-aligned data matrix
$\mathbf{U}, \mathbf{\Sigma}, \mathbf{V}$	—	SVD factors of $\mathbf{X}^{(e,c)}$
σ_k	$\mathbb{R}_{\geq 0}$	k-th singular value
$\mathbf{v}_k$	$\mathbb{R}^{\overline{W}}$	k-th right singular vector (basis shape)
$\mathbf{u}_k$	$\mathbb{R}^{n_e}$	k-th left singular vector (event loadings)
EVR_k	$[0, 1]$	Explained variance ratio of component k
SI	$\mathbb{R}_{\geq 0}$	Stereotypy index σ_1/σ_2
IC_k	$[-1, 1]$	Information Coefficient for component k
$r_i^{(H)}$	$\mathbb{R}$	Forward log-return at horizon H, event i
H	$H \in \mathbb{N}$	Forward-return horizon in candles
$s_j^{(e,c)}(t)$	$[-1, 1]$	Cosine-similarity score, channel c, event e
$S^{(e)}(t)$	$\mathbb{R}$	Multi-channel aggregated score for event e
w_c	$w_c \geq 0, \sum w_c = 1$	Channel weight for $c \in \mathcal{C}$
ATR_t	$\mathbb{R}_{>0}$	Average True Range at time t
ϕ_i	$[0, 1] \to \mathbb{R}$	Analytical basis function i
a	$a \in [3, 8]$	Exponential decay parameter

2 Introduction and Motivation

2.1 The Core Problem

Classical technical analysis assigns fixed shapes to named patterns. A "head and shoulders" has a specific geometry; a "hammer" has a long lower wick and a small body. These shapes are hand-designed by practitioners and carry two fundamental limitations: (1) the true shape of a pattern in a given market may differ from the textbook version, and (2) there is no principled mechanism for deciding *which* shape variation actually predicts future returns as opposed to merely being visually recognisable.

The convolver pipeline described in this document addresses both limitations. Events are defined *analytically* by price action conditions (Section 4), which is unambiguous and replicable. The shapes associated with those events are then *discovered from data* via Singular Value Decomposition (Section 6). Only the discovered shapes that survive an Information Coefficient gate (Section 7) are retained for deployment. The result is a kernel library that is both data-driven and statistically validated.

2.2 Analogues in Other Disciplines

The methodology is a direct analogue of two well-established signal processing paradigms.

Event-Related Potentials (ERP) in neuroscience. An ERP is computed by locking EEG recordings to a stimulus event and averaging across trials. The average converges to the signal phase-locked to the event, while random neural noise cancels. Our event-aligned matrix $\mathbf{X}^{(e,c)}$ plays exactly this role: each row is one "trial" (one occurrence of the event), and SVD generalises averaging by retaining not just the mean shape but the full low-dimensional subspace of shape variation. The left singular vector $\mathbf{u}_1$ encodes trial-to-trial amplitude variation; the right singular vector $\mathbf{v}_1$ encodes the dominant waveform shape.

Matched filtering in radar and seismology. A matched filter maximises the signal-to-noise ratio for detecting a known template $\mathbf{k}$ in a noisy signal $\mathbf{x}(t)$ by computing the cross-correlation $s(t) = \mathbf{k}^\top \mathbf{x}(t)$. The SVD right singular vectors $\mathbf{v}_k$ serve as data-driven templates, and the online scoring step (Section 9) is exactly a matched filter bank — one filter per surviving component.

2.3 Key Insight: Shape Discovery Before Return Fitting

The critical anti-overfitting property of this approach is the ordering of operations:

1. SVD discovers basis functions using only the *event-aligned price data*. It has no access to forward returns.
2. IC filtering checks which of these already-discovered shapes correlate with forward returns.

We are not fitting shapes to returns. We are asking whether naturally occurring shape variation — variation the market itself exhibits around analytically-defined events — is predictive. This ordering ensures that the discovered kernels are interpretable geometric objects, not overfit regression artefacts.

3 Candlestick Decomposition into Signal Channels

3.1 OHLCV Decomposition

A single OHLCV candle at time t is the tuple $(O_t, H_t, L_t, C_t, V_t)$ satisfying

$$L_t \le \min(O_t, C_t) \le \max(O_t, C_t) \le H_t, \quad V_t \ge 0. \tag{1}$$

We decompose each candle into four *signal channels*.

Definition 3.1 (Signal Channels)**.** For a candle at time t, define:

$$b_t := C_t - O_t \tag{2}$$
$$u_t := H_t - \max(C_t,\, O_t) \tag{3}$$
$$l_t := \min(C_t,\, O_t) - L_t \tag{4}$$
$$v_t := V_t \tag{5}$$

Remark 3.1. By (1), $u_t \ge 0$ and $l_t \ge 0$ always. The body b_t is signed: $b_t > 0$ for a bull candle (close above open), $b_t < 0$ for a bear candle. The decomposition is invertible: given (b_t, u_t, l_t, O_t) one recovers (O_t, H_t, L_t, C_t) exactly. The channel set is $\mathcal{C} = \{b, u, l, v\}$.

3.2 Window Extraction

Definition 3.2 (Event Window)**.** For an event detected at time t_e and window width $W \in \mathbb{N}$, the *event window* of channel $c \in \mathcal{C}$ is the vector

$$\mathbf{x}^{(c)}(t_e) := \left(x^{(c)}_{t_e-W+1},\ x^{(c)}_{t_e-W+2},\ \ldots,\ x^{(c)}_{t_e}\right)^\top \in \mathbb{R}^W, \tag{6}$$

where $x_t^{(c)} \in \{b_t, u_t, l_t, v_t\}$ for $c \in \{b, u, l, v\}$ respectively. The window uses $W-1$ *pre-event* candles plus the event candle itself. The within-window time index is $\tau \in \{0, 1, \ldots, W-1\}$, with $\tau = W-1$ corresponding to t_e.

3.3 Per-Window Normalisation

Raw price levels are not comparable across events separated in time or across asset regimes. We apply a *shape-preserving* normalisation that removes level and scale but retains geometry.

Definition 3.3 (ATR Normalisation)**.** Let ATR_t be the N_{atr}-period Average True Range at time t:

$$\mathrm{ATR}_t := \frac{1}{N_{\mathrm{atr}}} \sum_{j=0}^{N_{\mathrm{atr}}-1} \max\!\Big(H_{t-j} - L_{t-j},\ |H_{t-j} - C_{t-j-1}|,\ |L_{t-j} - C_{t-j-1}|\Big). \tag{7}$$

The normalised window for event i with event time $t_e^{(i)}$ is

$$\tilde{x}_i^{(c)}(\tau) := \frac{x^{(c)}_{t_e^{(i)}-W+1+\tau} - \mu_i^{(c)}}{\mathrm{ATR}_{t_e^{(i)}}}, \quad \tau = 0, 1, \ldots, W-1, \tag{8}$$

where $\mu_i^{(c)} := \frac{1}{W}\sum_{\tau=0}^{W-1} x^{(c)}_{t_e^{(i)}-W+1+\tau}$ is the within-window mean of channel c for event i.

Remark 3.2. For the unsigned channels (u, l, v), the sample standard deviation may be used instead of ATR as the denominator if preferred, with the caveat that near-zero standard deviations require a floor: $\max(\hat{\sigma}_i, \epsilon)$ for small $\epsilon > 0$. We recommend ATR because it is a global scale estimate robust to within-window outliers and consistent across channels.

4 Analytical Event Definitions

We define six event types in $\mathcal{E}$. Let N be the lookback period in candles and let ATR_t be as in (7).

4.1 Breakout Events

Definition 4.1 (Breakout, Bull)**.** A *bull breakout* occurs at time t if and only if

$$C_t > \max_{j=1}^{N} H_{t-j} \quad \text{and} \quad V_t > 1.5 \cdot \mathrm{median}\!\big(V_{t-N}, \ldots, V_{t-1}\big). \tag{9}$$

Definition 4.2 (Breakout, Bear)**.** A *bear breakout* occurs at time t if and only if

$$C_t < \min_{j=1}^{N} L_{t-j} \quad \text{and} \quad V_t > 1.5 \cdot \mathrm{median}\!\big(V_{t-N}, \ldots, V_{t-1}\big). \tag{10}$$

The volume condition filters genuine momentum moves from slow price drift: a real breakout typically registers at least $1.5\times$ the recent median volume. The N-period range maximum $\max_{j=1}^{N} H_{t-j}$ is equivalent to the upper Donchian channel (excluding the current bar). Both can be computed in $O(1)$ amortised time per bar using a monotone deque.

4.2 Turtle Soup Events (False Range Break)

Turtle Soup refers to a pattern described by Raschke and Connors (1995): price transiently penetrates a N-period extreme, then closes back inside the prior range within the same bar, suggesting that the breakout attracted stop-loss orders that were absorbed by a larger counterparty.

Definition 4.3 (Turtle Soup, Bull)**.** A *bull turtle soup* (fake breakdown, anticipates long) occurs at time t if and only if

$$L_t < \min_{j=1}^{N} L_{t-j} \quad \text{and} \quad C_t > \min_{j=1}^{N} L_{t-j}. \tag{11}$$

Definition 4.4 (Turtle Soup, Bear)**.** A *bear turtle soup* (fake breakout, anticipates short) occurs at time t if and only if

$$H_t > \max_{j=1}^{N} H_{t-j} \quad \text{and} \quad C_t < \max_{j=1}^{N} H_{t-j}. \tag{12}$$

In both cases the condition asserts that the candle's extreme pierced the historical range but the close reverted. The lower wick channel l_t is expected to be especially large for Definition 4.3 (the spike and reversal shows as a long lower wick), providing a signal-rich feature for SVD.

4.3 Trap Events (Confirmed False Breakout)

A trap event requires two-bar confirmation: the breakout must occur at time t, and the close at time $t + K$ must have reversed.

Definition 4.5 (Bull Trap)**.** A *bull trap* is a pair $(t, t + K)$ satisfying:

(i) Condition (9) holds at time t.

(ii) $C_{t+K} < C_t - \alpha \cdot \mathrm{ATR}_t$.

The event window is centred on the breakout candle at time t.

Definition 4.6 (Bear Trap)**.** A *bear trap* is a pair $(t, t + K)$ satisfying:

(i) Condition (10) holds at time t.

(ii) $C_{t+K} > C_t + \alpha \cdot \mathrm{ATR}_t$.

Remark 4.1 (Look-ahead bias)**.** Trap events (Definitions 4.5–4.6) use information from $t + K$ to label the event at t. They are therefore *only valid for offline kernel discovery.* The aligned window is extracted at t (pre-event candles plus the breakout candle), and the confirmation at $t+K$ is used solely to partition events into "trap" versus "genuine breakout" classes. Deploying the kernel in real time introduces no look-ahead, because the kernel is applied to the current window at bar t.

Parameter table. The event definitions involve the following configurable parameters:

Parameter	Symbol	Default	Valid Range
Range lookback	N	20 bars	$[10, 60]$
Volume multiplier	1.5	1.5	$[1.2, 2.5]$
ATR period	N_{atr}	14 bars	$[7, 28]$
Trap confirmation lag	K	3 bars	$[1, 10]$
ATR reversal threshold	α	1.0	$[0.5, 2.0]$

5 Event-Aligned Matrix Construction

Definition 5.1 (Event-Aligned Matrix)**.** Let $\mathcal{T}_e = \{t_e^{(1)}, t_e^{(2)}, \ldots, t_e^{(n_e)}\}$ be the ordered set of detection times for event type $e \in \mathcal{E}$. For channel $c \in \mathcal{C}$, the *event-aligned matrix* is

$$\mathbf{X}^{(e,c)} := \begin{pmatrix} \tilde{x}_1^{(c)}(0) & \tilde{x}_1^{(c)}(1) & \cdots & \tilde{x}_1^{(c)}(W-1) \\ \tilde{x}_2^{(c)}(0) & \tilde{x}_2^{(c)}(1) & \cdots & \tilde{x}_2^{(c)}(W-1) \\ \vdots & \vdots & \ddots & \vdots \\ \tilde{x}_{n_e}^{(c)}(0) & \tilde{x}_{n_e}^{(c)}(1) & \cdots & \tilde{x}_{n_e}^{(c)}(W-1) \end{pmatrix} \in \mathbb{R}^{n_e \times W}, \tag{13}$$

where $\tilde{x}_i^{(c)}(\tau)$ is given by (8).

Each row of $\mathbf{X}^{(e,c)}$ is one normalised event window. Each column corresponds to a relative time offset τ within the window. The matrix has the within-window mean removed by construction (8).

Optional column-wise centring. Some authors additionally remove the across-event mean at each time step:

$$\hat{X}_{i,\tau}^{(e,c)} \leftarrow \tilde{x}_{i,\tau}^{(c)} - \frac{1}{n_e} \sum_{i'=1}^{n_e} \tilde{x}_{i',\tau}^{(c)}. \tag{14}$$

Without (14): the first right singular vector $\mathbf{v}_1$ approximates the mean shape (preferred when the average pattern is itself informative). With (14): all components capture deviations from the mean pattern. We adopt the convention of *not* applying (14) as the default.

Minimum event count. We require

$$n_e \geq n_{\min} = 100 \tag{15}$$

before proceeding to decomposition. Below this threshold the singular vectors are unreliable estimators of the population shapes.

6 SVD Decomposition: Basis Discovery

6.1 The Decomposition

Definition 6.1 (Thin SVD)**.** For $\mathbf{X} := \mathbf{X}^{(e,c)} \in \mathbb{R}^{n_e \times W}$ with $n_e \geq W$, the *thin* (economy) Singular Value Decomposition is

$$\mathbf{X} = \mathbf{U}\,\mathbf{\Sigma}\,\mathbf{V}^\top, \tag{16}$$

where

$$\mathbf{U} \in \mathbb{R}^{n_e \times W}, \quad \mathbf{U}^\top \mathbf{U} = \mathbf{I}_W, \tag{17}$$

$$\mathbf{\Sigma} = \operatorname{diag}(\sigma_1, \sigma_2, \ldots, \sigma_W), \quad \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_W \geq 0, \tag{18}$$

$$\mathbf{V} \in \mathbb{R}^{W \times W}, \quad \mathbf{V}^\top \mathbf{V} = \mathbf{I}_W. \tag{19}$$

For $n_e < W$ the decomposition has rank n_e; in practice $n_e \gg W$ is guaranteed by (15).

The columns $\mathbf{v}_k = \mathbf{V}_{:,k} \in \mathbb{R}^W$ are the *discovered basis functions* (shapes), ordered from dominant to residual. The columns $\mathbf{u}_k = \mathbf{U}_{:,k} \in \mathbb{R}^{n_e}$ give the *per-event loading* on basis function k: the scalar $u_{ik} = U_{ik}$ measures how strongly event i projects onto shape k.

6.2 Interpretive Quantities

Definition 6.2 (Explained Variance Ratio)**.** The fraction of total squared variation explained by component k is

$$\mathrm{EVR}_k := \frac{\sigma_k^2}{\sum_{j=1}^{W} \sigma_j^2} = \frac{\sigma_k^2}{\|\mathbf{X}\|_F^2}, \tag{20}$$

where $\|\cdot\|_F$ denotes the Frobenius norm and the identity $\|\mathbf{X}\|_F^2 = \sum_j \sigma_j^2$ follows from the unitary invariance of the Frobenius norm.

Definition 6.3 (Stereotypy Index)**.** The *stereotypy index* of an event class (e, c) is

$$\mathrm{SI}^{(e,c)} := \frac{\sigma_1}{\sigma_2} \tag{21}$$

(with the convention SI $= \infty$ when $\sigma_2 = 0$). A high stereotypy index (SI $\gg 1$) indicates that a single dominant shape accounts for most of the variation: the event is *stereotyped*. An index near 1 indicates a broad, multi-modal distribution of shapes. We recommend requiring SI ≥ 1.5 before retaining components beyond $k = 1$.

6.3 Truncation Rule

We retain the top $K^{(e,c)}$ components where the cumulative explained variance first exceeds threshold $\rho^* = 0.80$:

$$K^{(e,c)} := \min\left\{ K \in \{1, \ldots, W\} \,:\, \sum_{k=1}^{K} \mathrm{EVR}_k \geq \rho^* \right\}, \quad \rho^* = 0.80. \tag{22}$$

The retained columns of $\mathbf{V}$ form the *candidate kernel set* $\mathcal{K}_{\mathrm{cand}}^{(e,c)} = \{\mathbf{v}_1, \ldots, \mathbf{v}_{K^{(e,c)}}\}$ before IC filtering.

6.4 Reconstruction Error

The best rank-K approximation to $\mathbf{X}$ in Frobenius norm is $\mathbf{U}_K \mathbf{\Sigma}_K \mathbf{V}_K^\top$, where subscript K denotes the leading K factors. The relative reconstruction error is

$$\epsilon_K := \frac{\left\|\mathbf{X} - \mathbf{U}_K \mathbf{\Sigma}_K \mathbf{V}_K^\top\right\|_F}{\|\mathbf{X}\|_F} = \sqrt{1 - \sum_{k=1}^{K} \mathrm{EVR}_k}. \tag{23}$$

By the Eckart–Young–Mirsky theorem, no rank-K matrix provides a smaller Frobenius-norm approximation to $\mathbf{X}$.

7 Predictive Filtering: The IC Gate

7.1 Forward Returns

Definition 7.1 (Forward Log-Return)**.** For event i at time $t_e^{(i)}$ and forward horizon H candles, the forward log-return is

$$r_i^{(H)} := \ln \frac{C_{t_e^{(i)}+H}}{C_{t_e^{(i)}}}. \tag{24}$$

This is the continuously compounded return from the event close to the close H bars later. The return begins strictly after the event window ends, so there is no overlap between the information used by SVD and the return being predicted.

7.2 Information Coefficient

Definition 7.2 (Information Coefficient, IC)**.** For SVD component k of event type (e, c), the Information Coefficient at horizon H is

$$\mathrm{IC}_k^{(e,c,H)} := \mathrm{corr}\big(\mathbf{u}_k,\ \mathbf{r}^{(H)}\big) = \frac{\sum_{i=1}^{n_e}\big(u_{ik} - \bar{u}_k\big)\big(r_i^{(H)} - \bar{r}^{(H)}\big)}{\sqrt{\sum_{i=1}^{n_e}\big(u_{ik} - \bar{u}_k\big)^2 \cdot \sum_{i=1}^{n_e}\big(r_i^{(H)} - \bar{r}^{(H)}\big)^2}}, \tag{25}$$

where $\bar{u}_k = n_e^{-1} \sum_i u_{ik}$ and $\bar{r}^{(H)} = n_e^{-1} \sum_i r_i^{(H)}$.

Component k is *retained* in the final kernel library if and only if

$$\big|\mathrm{IC}_k^{(e,c,H)}\big| \geq \delta_{\mathrm{IC}}, \qquad \delta_{\mathrm{IC}} = 0.03. \tag{26}$$

Remark 7.1 (Why this is not overfitting)**.** The loadings $\mathbf{u}_k = \mathbf{U}_{:,k}$ are determined entirely by the price-action matrix $\mathbf{X}^{(e,c)}$. The SVD objective minimises reconstruction error with no knowledge of $\mathbf{r}^{(H)}$. The IC computation is therefore a *test* on the SVD output, not a training step. This is analogous to computing the correlation between a PCA projection and an external label: the projection was not fitted to the label.

7.3 Multiple Comparison Correction

Across all combinations $(e, c, k, H) \in \mathcal{E} \times \mathcal{C} \times \{1, \ldots, K_{\max}\} \times \mathcal{H}$ (where $\mathcal{H}$ is the set of tested horizons), the IC tests constitute a family of hypotheses. We apply Benjamini–Hochberg FDR control at level $q = 0.05$:

1. For each test, compute the two-sided p-value under the null $H_0\colon \mathrm{IC} = 0$ using the t-statistic

$$t_k = \mathrm{IC}_k \sqrt{\frac{n_e - 2}{1 - \mathrm{IC}_k^2}}, \quad t_k \sim t_{n_e - 2} \text{ under } H_0. \tag{27}$$

2. Order p-values: $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(m)}$ for m total tests.
3. Reject H_0 for all tests with index $j \leq j^*$, where $j^* := \max\{j : p_{(j)} \leq jq/m\}$.

Only tests that survive BH-FDR correction *and* satisfy (26) are retained.

8 Kernel Construction from Discovered Components

8.1 Kernel Library

Definition 8.1 (Kernel Library)**.** After IC filtering, the *kernel library* is the set

$$\mathcal{K} := \big\{\mathbf{k}_j^{(e,c)} \in \mathbb{R}^W\ :\ e \in \mathcal{E},\ c \in \mathcal{C},\ j \text{ survives IC gate}\big\}, \tag{28}$$

where $\mathbf{k}_j^{(e,c)} := \mathbf{v}_j^{(e,c)}$ is the j-th right singular vector of $\mathbf{X}^{(e,c)}$ (Definition 6.1).

Proposition 8.1 (Kernel Properties)**.** Every kernel $\mathbf{k} \in \mathcal{K}$ satisfies:

(i) **Unit norm:** $\|\mathbf{k}\|_2 = 1$ (right singular vectors are orthonormal).

(ii) **Orthogonality within class:** for fixed (e, c), $\langle \mathbf{k}_j^{(e,c)}, \mathbf{k}_{j'}^{(e,c)} \rangle = 0$ for $j \neq j'$.

(iii) **Mean-subtracted:** kernels carry no level information by construction of normalisation (8).

Proof. Properties (i) and (ii) follow directly from the orthonormality of $\mathbf{V}$ in the SVD (16): $\mathbf{V}^\top\mathbf{V} = \mathbf{I}$. Property (iii) follows because the per-window mean $\mu_i^{(c)}$ is subtracted in (8) before forming $\mathbf{X}^{(e,c)}$. □

8.2 Directional Association

Each retained kernel $\mathbf{k}_j^{(e,c)}$ carries a signed IC. We record the predicted trading direction:

$$d_j^{(e,c)} := \operatorname{sign}\bigl(\mathrm{IC}_j^{(e,c,H^*)}\bigr), \tag{29}$$

where H^* is the primary forecast horizon. In production, $d_j^{(e,c)} \cdot S^{(e)}(t)$ is the signed signal: positive means long, negative means short.

9 Online Convolution Scoring

9.1 Per-Channel Score

At production time t, the current window for channel c is the normalised vector

$$\mathbf{x}_t^{(c)} := \bigl(\tilde{x}_{t-W+1}^{(c)}, \ldots, \tilde{x}_t^{(c)}\bigr)^\top \in \mathbb{R}^W, \tag{30}$$

normalised identically to (8) using ATR_t at the current bar. For kernel $\mathbf{k}_j^{(e,c)} \in \mathcal{K}$, the *cosine similarity score* is

$$s_j^{(e,c)}(t) := \frac{\langle \mathbf{x}_t^{(c)}, \mathbf{k}_j^{(e,c)} \rangle}{\bigl\|\mathbf{x}_t^{(c)}\bigr\|_2 \cdot \bigl\|\mathbf{k}_j^{(e,c)}\bigr\|_2} = \frac{\langle \mathbf{x}_t^{(c)}, \mathbf{k}_j^{(e,c)} \rangle}{\bigl\|\mathbf{x}_t^{(c)}\bigr\|_2}, \tag{31}$$

where the simplification uses $\|\mathbf{k}\|_2 = 1$ from Proposition 8.1. The score satisfies $s \in [-1, 1]$ by the Cauchy–Schwarz inequality, with $s = 1$ iff the current window is an exact positive scalar multiple of the kernel.

The inner product $\langle \mathbf{x}, \mathbf{k} \rangle = \sum_{\tau=0}^{W-1} x(\tau)k(\tau)$ is computed in $O(W)$ time per kernel per bar. This is the zero-lag output of a cross-correlation filter; the term "convolver" refers to the sliding evaluation of this inner product as new bars arrive.

9.2 Multi-Channel Aggregation

For event type e and component j, the multi-channel score is

$$S_j^{(e)}(t) := \sum_{c \in \mathcal{C}} w_c \cdot s_j^{(e,c)}(t), \quad \sum_{c \in \mathcal{C}} w_c = 1, \quad w_c \geq 0. \tag{32}$$

Equal-weight default: $w_c = 0.25$ for all $c \in \{b, u, l, v\}$. IC-weighted alternative:

$$w_c \propto \bigl|\mathrm{IC}_j^{(e,c,H^*)}\bigr|, \quad w_c \leftarrow w_c \,/\, \textstyle\sum_{c'} w_{c'}. \tag{33}$$

9.3 Composite Event Score

Aggregating across all surviving components j of event type e:

$$S^{(e)}(t) := \sum_{\{j \,:\, \mathbf{k}_j^{(e,\cdot)} \in \mathcal{K}\}} \mathrm{IC}_j \cdot S_j^{(e)}(t), \tag{34}$$

where the IC weight ensures that components with higher predictive power contribute more to the composite. The composite score $S^{(e)}(t) \in \mathbb{R}$ is the final feature output for event type e at time t, suitable for ingestion into the NAT feature pipeline.

10 Statistical Validation Framework

10.1 Walk-Forward Validation

Algorithm: Walk-Forward Kernel Discovery and Validation

1. Inputs: OHLCV dataset $\mathcal{D}$, split $\rho_{\text{train}} = 0.60$, horizon H.
2. Set $T_{\text{train}} := \lfloor \rho_{\text{train}} T \rfloor$.
3. **Discovery phase** on $t \leq T_{\text{train}}$:
  For each $(e, c) \in \mathcal{E} \times \mathcal{C}$:
    Detect events $\mathcal{T}_e$; build $\mathbf{X}^{(e,c)}$; compute SVD.
    Truncate to $K^{(e,c)}$ components via (22).
    Compute IC (25) using in-sample returns; apply gate (26).
4. Apply BH-FDR correction across all tests.
5. **Validation phase** on $t \in (T_{\text{train}}, T - H]$:
  For each out-of-sample event: score window with $\mathcal{K}$; record $S^{(e)}$ and $r^{(H)}$.
6. Compute out-of-sample IC for each surviving kernel.
7. Report: in-sample IC, out-of-sample IC, IC decay ratio (35).

IC Decay Ratio. A kernel is considered robust if

$$\rho_{\text{decay}} := \frac{\text{IC}_k^{\text{OOS}}}{\text{IC}_k^{\text{IS}}} \geq 0.5. \tag{35}$$

A ratio below 0.5 indicates in-sample overfitting.

10.2 Kernel Stability Across Rolling Windows

To test whether discovered shapes are stable over time (rather than artefacts of a specific regime), we perform a rolling re-discovery.

Definition 10.1 (Rolling SVD Stability)**.** For rolling windows $\mathcal{D}_1, \mathcal{D}_2, \ldots, \mathcal{D}_M$ of length T_{roll} with stride Δ, let $\mathbf{v}_1^{(m)}$ be the first right singular vector of $\mathbf{X}^{(e,c)}$ computed on $\mathcal{D}_m$. The *stability score* between consecutive windows is

$$\rho_m := \left|\langle \mathbf{v}_1^{(m)}, \mathbf{v}_1^{(m+1)} \rangle\right| = \left|\cos \angle(\mathbf{v}_1^{(m)}, \mathbf{v}_1^{(m+1)})\right|. \tag{36}$$

The absolute value is taken because SVD is unique only up to sign. The overall stability is $\bar{\rho} = (M-1)^{-1} \sum_{m=1}^{M-1} \rho_m$. We require $\bar{\rho} \geq 0.70$.

10.3 Minimum Event Count Justification

Recall condition (15): $n_e \geq 100$. Under a spiked covariance model, the variance of the estimated right singular vector $\hat{\mathbf{v}}_k$ is of order $O(\sigma^2 W / (n_e \cdot \sigma_k^2))$. For typical values ($W = 20$, $\sigma^2/\sigma_k^2 \approx 10$), we need $n_e \gtrsim 50$ to achieve $O(10\%)$ estimation error, and $n_e = 100$ provides a comfortable margin.

10.4 Sensitivity Analysis

Re-run the full pipeline at parameter perturbations $\{N \pm 5,\ K \pm 1,\ \alpha \pm 0.25\}$ and verify:

(i) Surviving kernels at base parameters also survive at $\geq 3/4$ of perturbed configurations.

(ii) IC values are stable: $\Delta\text{IC} < 0.01$ across perturbations.

11 Integration with the Existing Feature Pipeline

11.1 Feature Augmentation

The NAT ingestor produces 209 features per 100 ms emission. The convolver scores $S^{(e)}(t)$ for $e \in \mathcal{E}$ form $|\mathcal{E}| = 6$ additional scalar features. Including per-component scores, the maximum expansion is $6 \times 4 \times K_{\max}$ features before IC filtering; in practice the gate reduces this substantially.

The convolver features operate at candle granularity (e.g. 1-minute or 5-minute bars), whereas the base 209 features operate at 100 ms. Alignment is achieved by repeating the candle-level score for all sub-candle emissions belonging to that bar.

11.2 Expected Decorrelation

The convolver captures multi-candle *shape* over a W-bar window. The existing imbalance cluster (`imbalance_qty_l1`, `imbalance_qty_l5`, etc.) captures instantaneous order flow at the quote level. These are structurally distinct: imbalance features are point-in-time, ∼100 ms resolution, order book state; convolver features are window-averaged, W-candle resolution, price geometry. We expect Pearson correlation $|r(\text{convolver}, \text{imbalance})| < 0.3$ in typical conditions, providing genuine diversification in the alpha stack.

11.3 Regime Gating

The existing entropy features (`ent_permutation_entropy`, `ent_approximate_entropy`) measure the regularity of the price process. In low-entropy regimes the price process is more deterministic and event patterns are more stereotyped (higher $\mathrm{SI}^{(e,c)}$). We recommend the regime gate:

$$S^{(e)}_{\text{gated}}(t) := S^{(e)}(t) \cdot \mathbf{1}\big[\text{PermEnt}_t < \eta_{\text{thresh}}\big], \tag{37}$$

where $\eta_{\text{thresh}} \in (0, 1)$ is calibrated on the training period. This suppresses convolver signals during high-entropy (random-walk-like) regimes where pattern-based predictions are unreliable.

12 Basis Function Initialisation: Analytical Warm-Start

12.1 Motivation

The SVD discovers basis functions purely from data. As a diagnostic and optional initialisation, we define an analytical basis dictionary against which data-driven shapes can be compared. This serves two purposes:

1. **Interpretability**: if $\mathbf{v}_1$ has high cosine similarity to a known basis function, the event shape admits a classical description.
2. **Warm-start**: when data is limited ($n_e < n_{\min}$), analytical bases can substitute for SVD-derived ones, at the cost of reduced adaptivity.

12.2 Analytical Basis Dictionary

Let $\bar{\tau} = \tau/(W - 1) \in [0, 1]$ be the normalised within-window coordinate, with $\bar{\tau} = 0$ at the start of the window and $\bar{\tau} = 1$ at the event candle.

Polynomial family (low-frequency trend shapes).

$$\phi_{q1}(\bar{\tau}) = \bar{\tau}^2 \tag{38}$$

$$\phi_{q2}(\bar{\tau}) = \left(\bar{\tau} - \tfrac{1}{2}\right)^2 \tag{39}$$

$$\phi_{q3}(\bar{\tau}) = (1 - \bar{\tau})^2 \tag{40}$$

These capture convex and concave acceleration patterns: ϕ_{q1} is convex-up (acceleration toward the event), ϕ_{q3} is convex-down (deceleration into the event), and ϕ_{q2} is a U-shape (dip then recovery, or vice versa).

Cubic family (S-curves and reversal structures).

$$\phi_{c1}(\bar{\tau}) = \bar{\tau}^3 \tag{41}$$

$$\phi_{c2}(\bar{\tau}) = \left(\bar{\tau} - \tfrac{1}{2}\right)^3 \tag{42}$$

$$\phi_{c3}(\bar{\tau}) = (1 - \bar{\tau})^3 \tag{43}$$

$$\phi_{s1}(\bar{\tau}) = 3\bar{\tau}^2 - 2\bar{\tau}^3 \tag{44}$$

The smoothstep ϕ_{s1} (Hermite easing function) satisfies $\phi_{s1}(0) = 0$, $\phi_{s1}(1) = 1$, $\phi'_{s1}(0) = \phi'_{s1}(1) = 0$, making it a natural model for a gradual acceleration-then-deceleration run-up into a breakout.

Exponential family (spike and impulse shapes). For $a \in [3, 8]$:

$$\phi_{e1}(\bar{\tau}; a) = e^{-a\bar{\tau}} \tag{45}$$

$$\phi_{e2}(\bar{\tau}; a) = e^{-a(1-\bar{\tau})} \tag{46}$$

$$\phi_{b1}(\bar{\tau}; a) = \exp\!\left(-a(\bar{\tau} - \tfrac{1}{2})^2\right) \tag{47}$$

ϕ_{e1} models a sharp spike at $\bar{\tau} = 0$ decaying to the event; ϕ_{e2} models a ramp-up climaxing at the event; ϕ_{b1} models a Gaussian bump centred at the window midpoint (useful for detecting the volume bulge that accompanies a breakout).

12.3 General Parametric Kernel

A general kernel from this dictionary is the normalised linear combination:

$$k(\bar{\tau}; \boldsymbol{\theta}) := \frac{\displaystyle\sum_i \theta_i \phi_i(\bar{\tau})}{\left\| \displaystyle\sum_i \theta_i \phi_i \right\|_2}, \quad \boldsymbol{\theta} \in \mathbb{R}^{|\Phi|}, \tag{48}$$

where $\Phi = \{\phi_{q1}, \phi_{q2}, \phi_{q3}, \phi_{c1}, \phi_{c2}, \phi_{c3}, \phi_{s1}, \phi_{e1}, \phi_{e2}, \phi_{b1}\}$ is the full basis set (10 elements).

12.4 Alignment Diagnostic

For each SVD-discovered basis vector $\mathbf{v}_k^{(e,c)}$ and each analytical basis ϕ_i (discretised to the W-point grid), the alignment is

$$\rho_{k,i}^{(e,c)} := \left| \langle \mathbf{v}_k^{(e,c)}, \hat{\boldsymbol{\phi}}_i \rangle \right|, \tag{49}$$

where $\hat{\boldsymbol{\phi}}_i = \boldsymbol{\phi}_i / \|\boldsymbol{\phi}_i\|_2$ is the normalised discrete version of ϕ_i. Values of $\rho > 0.85$ indicate strong alignment (the data-driven shape is interpretable as the corresponding analytical form). Values of $\rho < 0.5$ indicate the market produces a shape not captured by the analytical dictionary — the regime where data-driven discovery adds the most value.

13 Computational Complexity

13.1 Offline Discovery Phase

Step	Operation	Complexity	Notes
Event detection	Sliding max/min over N bars	$O(T)$ per type	Monotone deque
Window extraction	Copy $n_e \times W$ values	$O(n_e W)$	Single pass
Normalisation	Per-window mean and ATR	$O(n_e W)$	
SVD (thin)	LAPACK `dgesdd`	$O(n_e W^2)$	$n_e \gg W$ assumed
IC computation	Correlation of length n_e	$O(n_e)$	Per component
BH-FDR	Sort m p-values	$O(m \log m)$	$m \ll n_e$ typically

The dominant step is SVD: $O(n_e W^2)$ per (e, c) pair, giving total offline complexity $O(|\mathcal{E}| \cdot |\mathcal{C}| \cdot n_e W^2)$. For typical values ($|\mathcal{E}| = 6$, $|\mathcal{C}| = 4$, $n_e = 500$, $W = 20$), this is approximately $6 \times 4 \times 500 \times 400 = 4.8 \times 10^6$ floating-point operations — sub-second on any modern CPU.

13.2 Online Scoring Phase

For each new candle, the convolver evaluates:

$$\text{Cost per candle} = O\big(|\mathcal{K}| \cdot W\big), \tag{50}$$

where $|\mathcal{K}|$ is the number of retained kernels (typically ≤ 20 after IC filtering) and W is the window width. For $|\mathcal{K}| = 20$ and $W = 20$, this is 400 multiply-accumulate operations per candle, well within real-time constraints at any candle frequency.

13.3 Memory Requirements

The kernel library requires $|\mathcal{K}| \times W$ floats. For $|\mathcal{K}| = 20$ and $W = 20$ with 64-bit floats: $20 \times 20 \times 8 = 3.2\,\text{KB}$. The event-aligned matrices require $|\mathcal{E}| \times |\mathcal{C}| \times n_e \times W$ floats during discovery: $6 \times 4 \times 500 \times 20 \times 8 = 1.92\,\text{MB}$.

14 Summary: Pipeline Overview

The complete pipeline from raw OHLCV data to deployed convolver features:

1. **Candle decomposition** (Section 3): map each $(O_t, H_t, L_t, C_t, V_t)$ to channels (b_t, u_t, l_t, v_t) via (2)–(5), then normalise via (8).
2. **Event detection** (Section 4): apply analytical conditions (9)–(**??**) to label each bar as belonging to one of $|\mathcal{E}| = 6$ event types.
3. **Matrix assembly** (Section 5): for each (e, c) pair, stack normalised W-bar windows into $\mathbf{X}^{(e,c)} \in \mathbb{R}^{n_e \times W}$ (13).
4. **SVD basis discovery** (Section 6): factorise each matrix via (16); truncate to the top $K^{(e,c)}$ components explaining $\geq 80\%$ of variance (22).
5. **IC gate** (Section 7): correlate per-event SVD loadings $\mathbf{u}_k$ with forward returns $\mathbf{r}^{(H)}$ via (25); apply BH-FDR and $|\text{IC}| \geq 0.03$ threshold (26).
6. **Kernel library** (Section 8): surviving right singular vectors form $\mathcal{K}$ (28).
7. **Online scoring** (Section 9): at each bar, evaluate cosine similarities (31), aggregate across channels (32), weight by IC (34).

8. **Validation** (Section 10): walk-forward OOS IC, rolling stability $\bar{\rho} \geq 0.70$ (36), IC decay ratio ≥ 0.50 (35).

9. **Integration** (Section 11): append convolver scores to NAT feature vector; apply regime gate (37).

Anti-overfitting guarantees. The pipeline provides three independent layers of protection against spurious discoveries:

(i) SVD is return-blind: shapes are discovered without access to forward returns.

(ii) BH-FDR controls the expected fraction of false discoveries at $q = 5\%$ across all tested (e, c, k, H) combinations.

(iii) Walk-forward OOS validation tests generalisability to unseen data (Sections 10).

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